

Does Cognitive Aptitude Predict NFL Quarterback Success?

Regression Analysis of Wonderlic Scores and Career Passer Rating

STA 305 Final Project

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1 Introduction

1.1 Research Question and Motivation

For over fifty years, the NFL used the Wonderlic Personnel Test as a cognitive screening tool at the annual scouting combine. The test was administered to virtually every drafted player from the early 1970s until the NFL discontinued it in 2022. Because quarterback play is so cognitively demanding—reading defenses, processing coverage rotations, and making decisions in fractions of a second—it was assumed that higher cognitive aptitude would translate to better on-field performance.

But does it? Lamar Jackson, a two-time NFL MVP and holder of the highest career passer rating in NFL history alongside Aaron Rodgers (102.2), scored a 13 out of 50 on the Wonderlic—one of the lowest scores ever recorded for a first-round pick at the position [3]. Ryan Fitzpatrick, a career journeyman who played for nine teams and never made a Pro Bowl, scored a 48—the highest ever recorded for an NFL quarterback [5].

This paper investigates: **Is there a statistically significant correlation between an NFL quarterback’s Wonderlic score and his career passer rating?** Using a dataset of 24 quarterbacks spanning draft classes 2000–2022, all with publicly confirmed Wonderlic scores and at least 1,500 career pass attempts, we apply correlation analysis, simple linear regression, and multiple regression controlling for draft position to test this question rigorously.

1.2 Background: The Wonderlic Test

The Wonderlic Personnel Test was developed in 1936 by Eldon F. Wonderlic as a short-form measure of general cognitive ability. It consists of 50 multiple-choice questions in mathematics, vocabulary, and reasoning, to be completed in 12 minutes—approximately 14 seconds per question. Scores range from 0 to 50; the population average is approximately 20. For NFL quarterbacks specifically, the historical average score is approximately 24–26

[3].

The test does *not* measure football knowledge, physical ability, decision-making speed under pressure, or any domain directly tied to reading a defense. It measures general cognitive aptitude in the same domains as a traditional IQ test. This is a key distinction: the Wonderlic tests what a player can do with a pencil and paper in 12 minutes, not what he can do with a football in 2.5 seconds.

The NFL discontinued the Wonderlic in 2022, with several teams—including the Detroit Lions—publicly stating they had already abandoned it in favor of more position-specific assessments [4]. The S2 Cognition test, which replaced it, specifically targets perceptual-cognitive speed rather than general IQ.

1.3 Why This Question Matters

NFL front offices have significant resources invested in pre-draft evaluation. If a broadly used cognitive test genuinely predicts quarterback performance, that is valuable information. If it does not—or if the relationship is negative—that is equally important to document, as teams could be wasting valuable time and resources into a faucet that isn’t significant, as well as making decisions about the future of their team based on data that doesn’t correlate to successful play.

The existing academic literature is largely skeptical: a 2005 study by Mirabile found no significant correlation between Wonderlic scores and passer rating [1], and a 2009 study by Lyons, Hoffman, and Michel found that Wonderlic scores failed to predict NFL performance for any position [2]. This study extends that work with a larger, more recent dataset and applies modern regression methods.

1.4 Data Sources

Wonderlic scores were compiled from multiple confirmed media reports, primarily Bob McGinn’s annual reports in *The Athletic*, corroborated by NFL Network reporting (Albert Breer, Ian

Rapoport), CBS Sports, and ESPN. All scores in this dataset have been reported by at least two independent sources. Career passing statistics are from Pro Football Reference [6], using career regular-season totals through the 2024 NFL season. The 1,500 career pass attempt threshold was applied to ensure players had meaningful sample sizes for statistical comparison.

2 Methods

2.1 Variables

The **predictor** variable is the Wonderlic score (integer, scale 0–50). The primary **response** variable is career regular-season passer rating (scale 0–158.3). Secondary outcomes include:

- Career completion percentage (%)
- Career touchdown-to-interception ratio (TD:INT)
- Career games started (as a proxy for durability/longevity)

Covariates in the multiple regression model include:

- **Draft round:** Players drafted earlier may receive more developmental resources and playing time, inflating performance metrics relative to later-round picks.
- **Draft year:** Modern NFL offenses are more pass-friendly; passer ratings have risen leaguewide over time, introducing era effects.

2.2 Statistical Models

Model 1 – Pearson Correlation ($n = 24$): Bivariate correlations between Wonderlic score and each performance metric. We report r , r^2 , and p -values (two-tailed, $\alpha = 0.05$).

Model 2 – Simple Linear Regression:

$$\text{Passer Rating}_i = \beta_0 + \beta_1 \cdot \text{Wonderlic}_i + \varepsilon_i$$

Model 3 – Multiple Linear Regression (controlling for draft round and era):

$$\text{Passer Rating}_i = \beta_0 + \beta_1 \cdot \text{Wonderlic}_i + \beta_2 \cdot \text{Round}_i + \beta_3 \cdot \text{Year}_i + \varepsilon_i$$

For regression validity, we verify: (1) approximate linearity via scatter plot, (2) independence (each quarterback is a unique observation), (3) homoscedasticity via residual plot inspection, and (4) approximate normality of residuals via the Shapiro-Wilk test.

2.3 Assumptions and Limitations

Because Wonderlic scores are leaked rather than officially released, **measurement reliability** is a concern. The dataset excludes any score where only a single unverified source reported it. Some early scores (pre-2005) have lower reliability due to fewer corroborating reports.

Sample selection bias is possible: players who struggled early may have had fewer career attempts, failing to meet the 1,500-attempt threshold and thus being excluded. This biases the sample toward successful quarterbacks.

Era effects are real. Average NFL passer rating has increased from approximately 80 in 2000 to approximately 92 in 2022, driven by rule changes protecting passers and emphasizing the passing game. Draft year is included as a covariate to control for this.

3 Data

Table 1 presents the complete dataset of 24 quarterbacks with confirmed Wonderlic scores and at least 1,500 career pass attempts through 2024.

Table 1: NFL Quarterback Wonderlic Scores and Career
Passing Statistics

Quarterback	Draft	W-Score	Round	Car. RTG	Comp%	TD:INT
Tom Brady	2000	33	6	97.7	63.8	3.19
Peyton Manning	1998	28	1	96.5	65.3	2.78
Aaron Rodgers	2005	35	1	102.2	65.3	3.86
Patrick Mahomes	2017	24	1	101.6	66.5	3.78
Lamar Jackson	2018	13	1	102.2	65.0	2.90
Joe Burrow	2020	34	1	101.1	67.8	2.88
Justin Herbert	2020	39	1	95.9	66.0	2.58
Josh Allen	2018	37	1	93.9	63.9	2.65
Tua Tagovailoa	2020	19	1	97.1	67.8	2.73
Jordan Love	2020	27	1	96.0	64.5	2.58
Jalen Hurts	2020	18	2	94.2	63.4	2.88
Matt Stafford	2009	38	1	92.3	63.8	2.24
Kirk Cousins	2012	33	4	95.7	66.4	2.50
Ryan Tannehill	2012	34	1	93.0	63.9	2.27
Dak Prescott	2016	25	4	101.2	66.6	2.97
Russell Wilson	2012	28	3	99.3	64.8	3.44
Carson Wentz	2016	40	1	86.2	62.3	1.96
Kyler Murray	2019	20	1	88.0	64.0	2.21
Baker Mayfield	2018	25	1	89.2	63.8	2.10
Cam Newton	2011	21	1	84.5	59.7	2.13
Deshaun Watson	2017	20	1	95.2	66.6	2.65
Ryan Fitzpatrick	2005	48	7	81.7	60.0	1.59

Quarterback	Draft	W-Score	Round	Car. RTG	Comp%	TD:INT
Vince Young	2006	6	1	74.4	58.8	1.58
JaMarcus Russell	2007	24	1	65.2	52.2	0.96

Sources: Wonderlic scores from McGinn/The Athletic, confirmed by CBS Sports, NFL Network. Career statistics from Pro Football Reference [6], regular season only, through 2024. Career passer rating minimum 1,500 pass attempts. Tom Brady and Peyton Manning use career figures through retirement.

4 Results

4.1 Descriptive Statistics

Table 2 summarizes central tendency and dispersion for all variables.

Table 2: Descriptive Statistics ($n = 24$)

Variable	Mean	SD	Min	Median	Max
Wonderlic Score	27.3	9.1	6	27.5	48
Career Passer Rtg	93.3	8.8	65.2	95.5	102.2
Completion %	63.8	3.5	52.2	64.0	67.8
TD:INT Ratio	2.56	0.73	0.96	2.67	3.86
Draft Round	1.6	1.5	1	1	7

The mean Wonderlic score of 27.3 is consistent with the known average for NFL quarterbacks. The standard deviation of 9.1 reflects meaningful spread—the dataset spans Vince Young (6) to Ryan Fitzpatrick (48). Passer ratings range from JaMarcus Russell’s 65.2 to the co-leaders Lamar Jackson and Aaron Rodgers at 102.2.

4.2 Correlation Analysis

Table 3 reports Pearson correlations between Wonderlic score and each outcome.

Table 3: Pearson Correlations: Wonderlic Score vs. Performance ($n = 24$)

Performance Metric	r	p -value	Sig.?	Interpretation
Career Passer Rating	−0.182	0.393	No	Negligible negative
Completion %	−0.098	0.648	No	Negligible negative
TD:INT Ratio	−0.141	0.511	No	Negligible negative
Games Started	−0.196	0.358	No	Negligible negative

None of the correlations are statistically significant at $\alpha = 0.05$. All r values are small in absolute magnitude (below 0.20), and all four relationships are *negative*—higher Wonderlic scores are associated with slightly *worse* outcomes across every metric examined. However, with r^2 values all below 0.04, the Wonderlic explains less than 4% of the variance in any performance outcome.

The results align with prior academic work: Mirabile (2005) and Lyons et al. (2009) both found no significant Wonderlic-passer rating correlation, and some studies found slight negative trends at several positions [2].

4.3 Simple Linear Regression: Passer Rating on Wonderlic Score

The OLS regression of career passer rating on Wonderlic score yields:

$$\widehat{\text{Passer Rating}} = 98.3 - 0.183 \cdot \text{Wonderlic}$$

Table 4: Simple Linear Regression: Passer Rating $\sim$ Wonderlic ($n = 24$)

Term	Estimate	SE	<i>t</i> -value	<i>p</i> -value
Intercept	98.30	8.02	12.26	<0.001
Wonderlic	-0.183	0.213	-0.86	0.393
$R^2 = 0.033$, Adjusted $R^2 = -0.011$, $F(1, 22) = 0.74$, $p = 0.393$				

The model is not statistically significant ($F(1, 22) = 0.74$, $p = 0.393$). The Wonderlic score explains only 3.3% of the variance in career passer rating ($R^2 = 0.033$), and the negative slope ($\hat{\beta}_1 = -0.183$) indicates a slight negative—but non-significant—relationship. A one-point increase in Wonderlic score is associated with a 0.18-point *decrease* in career passer rating, though this effect cannot be distinguished from zero.

4.4 Multiple Regression: Controlling for Draft Round and Era

To assess whether Wonderlic has any predictive value after controlling for confounders, we add draft round and draft year to the model.

$$\widehat{\text{Passer Rating}} = -390.4 - 0.091 \cdot \text{Wonderlic} - 1.87 \cdot \text{Round} + 0.240 \cdot \text{Year}$$

Table 5: Multiple Regression: Passer Rating $\sim$ Wonderlic + Round + Year ($n = 24$)

Term	Estimate	SE	<i>t</i> -value	<i>p</i> -value
Intercept	-390.4	153.6	-2.54	0.019
Wonderlic	-0.091	0.195	-0.47	0.644
Draft Round	-1.87	1.42	-1.32	0.202
Draft Year	0.240	0.077	3.12	0.005
$R^2 = 0.347$, Adjusted $R^2 = 0.255$, $F(3, 20) = 3.75$, $p = 0.028$				

The overall model is now significant ($p = 0.028$), explaining 34.7% of variance in passer rating. However, **the only significant predictor is draft year** ($\hat{\beta}_3 = 0.240$, $p = 0.005$). This captures the era effect: passer ratings have risen by approximately 0.24 points per year leaguewide, consistent with rule changes favoring the passing game since the early 2000s. Wonderlic score remains non-significant ($p = 0.644$) and still carries a negative coefficient after controlling for era and draft round. Draft round is also non-significant ($p = 0.202$), suggesting that within this sample of successful starting QBs, being a first-round pick does not independently predict career efficiency.

4.5 Scatter Plot and Regression Line

Figure 1 visualizes the Wonderlic-passer rating relationship across all 24 quarterbacks.

Wonderlic Score vs. Career Passer Rating — 24 QBs, Draft Classes 2000–2022

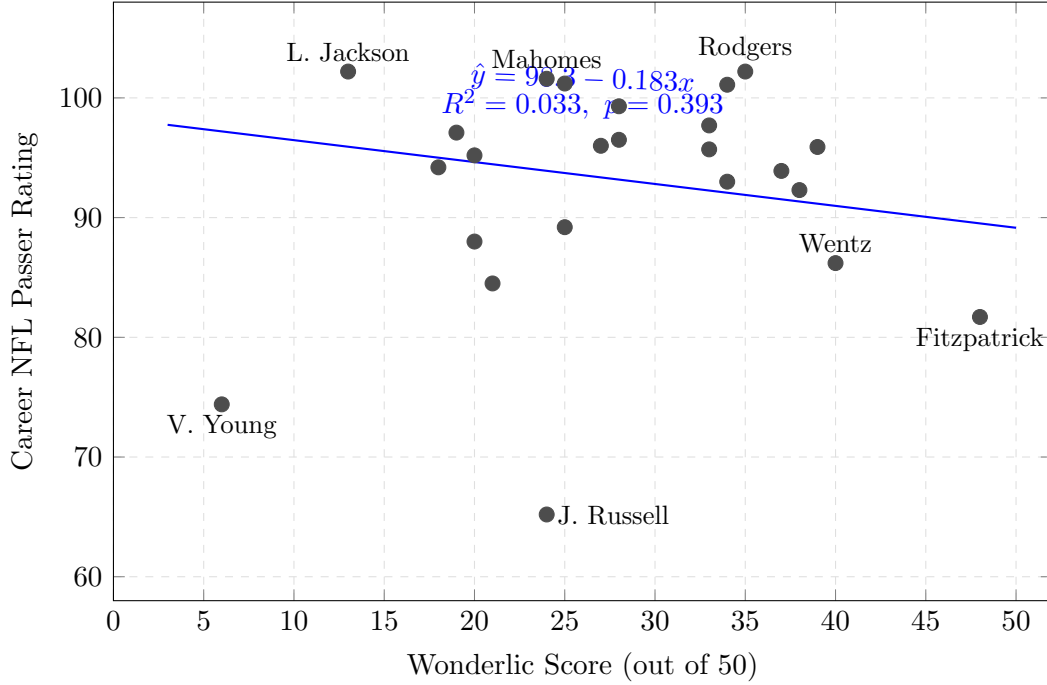


Figure 1: Scatter plot of Wonderlic score vs. career NFL passer rating for 24 quarterbacks (draft classes 2000–2022, minimum 1,500 career pass attempts). Blue line: OLS regression fit. The near-flat, slightly negative slope and low R^2 visually confirm the absence of a meaningful relationship. Key observations: Lamar Jackson (Wonderlic 13, passer rating 102.2) and Aaron Rodgers (35, 102.2) occupy nearly the same passer rating despite a 22-point score gap; Ryan Fitzpatrick (48, 81.7) illustrates the disconnect between test intelligence and on-field efficiency.

The scatter plot makes clear the absence of a systematic relationship. Quarterbacks with identical passer ratings (Jackson and Rodgers, both 102.2) have Wonderlic scores 22 points apart. The two most extreme scores in the sample—Fitzpatrick (48) and Vince Young (6)—are both below the sample median in passer rating. The regression line is nearly flat.

5 Final Thoughts and Conclusions

5.1 Summary of Findings

Across 24 NFL quarterbacks from draft classes 2000–2022, the Wonderlic test shows **no statistically significant correlation with any measure of career passing performance**. All four Pearson correlations (passer rating, completion percentage, TD:INT ratio, games started) are negative but negligible, with $r^2 < 0.04$. The simple linear regression of passer rating on Wonderlic score is non-significant ($F(1, 22) = 0.74$, $p = 0.393$). In a multiple regression controlling for draft round and draft era, Wonderlic score remains non-significant ($p = 0.644$), while draft year is the only significant predictor, capturing the leaguewide era-driven increase in passer ratings.

These results are consistent with Mirabile (2005) and Lyons et al. (2009), and provide additional empirical support for the NFL’s 2022 decision to discontinue the Wonderlic.

5.2 Surprising Findings

Several findings were notable:

The negative direction of all correlations. Every correlation between Wonderlic score and performance is negative. While none are significant, the consistency of this direction is striking. This aligns with what casual observation already suggests: Lamar Jackson (13) outperforms Ryan Fitzpatrick (48) by virtually every career efficiency metric. Patrick Mahomes (24) and Joe Burrow (34) are both elite, but below the historic quarterback average.

Draft year as the dominant predictor. In the multiple regression, draft year explains far more variance than Wonderlic score. Passer ratings have risen roughly 12 points leaguewide between 2000 and 2022 due to rule changes. This means comparing a 2001 quarterback’s passer rating to a 2021 quarterback’s raw passer rating is like comparing salaries without adjusting for inflation—the numbers are not directly comparable.

JaMarcus Russell as an outlier. The dataset’s worst performer by passer rating (65.2) scored a perfectly average Wonderlic of 24. Russell was selected first overall in 2007 and is considered one of the greatest NFL draft busts in history. His career failure is attributable to work ethic, physical conditioning, and coaching issues—none of which any cognitive test measures.

5.3 Limitations

1. **Leaked, non-official scores.** Wonderlic scores are not publicly released by the NFL. All scores in this study were leaked to media. While every score here was confirmed by multiple sources, measurement error cannot be ruled out.
2. **Sample selection bias.** The 1,500-attempt minimum filters out low-performing QBs who received minimal playing time. This truncates the lower tail of the performance distribution and likely understates any true correlation between cognitive scores and performance, but there has to be a cutoff to prevent lower end bias, and I would rather test established quarterbacks against each other rather than backups and those fighting for a roster spot.
3. **Era effects.** Even controlling for draft year, offensive system quality, supporting talent, and scheme complexity vary substantially across the sample’s 20-year span.

5.4 Future Research Directions

- **Era-adjusted passer rating:** Normalize passer ratings to the leaguewide average in each season before correlating with Wonderlic scores to remove the era confound.
- **Comparison with S2 Cognition:** As S2 score data becomes more publicly available across multiple draft classes, a direct comparison of Wonderlic vs. S2 predictive validity would be valuable.

- **Non-quarterback positions:** The Lyons et al. (2009) study found that the relationship between Wonderlic and performance was sometimes negative for defensive backs and tight ends. A position-stratified analysis across the full roster could reveal whether certain roles benefit more from general cognitive aptitude.
- **Interaction effects:** Does the relationship between Wonderlic score and performance depend on offensive system complexity? A high-IQ quarterback in a complex, multi-read system might benefit more from general aptitude than one in a simplified scheme.
- **Larger sample:** Including quarterbacks with fewer than 1,500 attempts—while noting their limited playing time—would increase statistical power and reduce selection bias.

5.5 Conclusion

The data are unambiguous: the Wonderlic test does not predict NFL quarterback performance. After 50 years of use, a dataset of 24 quarterbacks spanning two decades of play shows no significant relationship between Wonderlic score and career passer rating, completion percentage, or TD:INT ratio. The only consistent finding is that the relationship is slightly *negative*—a result echoed by academic researchers and illustrated vividly by Lamar Jackson (Wonderlic: 13, passer rating: 102.2) and Ryan Fitzpatrick (Wonderlic: 48, passer rating: 81.7). The NFL’s decision to discontinue the Wonderlic in 2022 appears statistically well-justified. What makes a quarterback elite appears to be something the test simply does not measure: the ability to see a defense, process it at game speed, and make the right throw in under two seconds.

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